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# Deliverable D2
# Workpackage WP.1
# Analysis of suitable statistical methods

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In the initial phase of the project, we conducted a survey comparing distributions. We focused on the basic properties of statistical functions for comparing distributions as well as applications in machine learning. We focused on methods for comparing distributions, which can then be used to compare weight distributions before and after the quantization process of a neural model in order to reduce the complexity of the neural model.

Based on this, we identified the following statistical methods:

a) Kullback-Leibler divergence

b) Jensen-Shannon divergence

c) Cosine similarity

Each method was evaluated according to the criteria:

- Sensitivity: how sensitively it can detect small changes in the distribution of weights.
- Computational efficiency: time and memory requirements of computation in the context of experiments with large models.
- Interpretability: The ease of interpreting the resulting numberAj keď niektoré metódy sa môžu zdať viac náročné na výpočet pri prvých testoch sa metódy väčším nárokom na výpočet dokázali poskytnúť viac informácií a prejavili schopnosť citlivejšie reagovať na malé zmeny.

### a) Kullback-Leiblerova divergence (KL divergence)

KL divergence is a statistical method used to quantify the difference between two probability distributions. It measures how much information is lost when one distribution serves as an approximation of the other. This property makes it potentially useful in fields such as machine learning, information theory, and data analysis.

Mathematical definition:
For two probability distributions P and Q, defined as:

$$D\_KL(P \mid\mid Q) = sum\_(i=1)\hat{}n \; P(i) * log(\, P(i) \,/\, Q(i)\,)$$

The KL divergence result is always non-negative, whereby:

- A value of 0 indicates a complete matching of distributions (no difference),
- A higher value indicates a greater divergence between the distributions.

Kullback-Leibler divergence is a fundamental information-theoretic metric that quantifies the difference between two probability distributions [1]. It is also referred to as relative entropy. KL divergence measures how much additional information is required to encode data from one distribution using code created for another distribution. This asymptotic measure provides information about how the distribution being compared differs from the reference distribution.

In the field of machine learning, KL divergence has important applications in a variety of disciplines. In training generative models such as variational autoencoders (VAEs), it acts as a regularization feature that ensures that the latent representations converge to a predefined distribution [4]. In the context of neural network compression, the assumption is that it could quantify the loss of information when quantizing weights or activations. In knowledge distillation, it is used to measure the difference between the output distributions of the teacher and student models. In Bayesian inference, it quantifies the difference between a priori and posterior distributions, while in information theory it quantifies the inefficiency of using the wrong probabilistic model [2], [9].

Mathematically, the KL divergence for two discrete probability distributions P and Q is defined as the sum of the products of the probabilities from the distribution P and the logarithms of the probability ratio P/Q. For continuous distributions, the summation is replaced by the integral. The resulting value is always nonnegative, with a zero value indicating identical distributions [1]. The higher the KL value of the divergence, the greater the difference between the distributions. An important property is asymmetry - KL(P||Q) is generally not equal to KL(Q||P), which means that it depends on which distribution we consider as the reference.

In the case of neural network quantization, KL divergence has the potential to indicate how quantization changes the distribution of model weights. Before quantization, the weights typically have a continuous distribution with floating-point values, but after quantization they are restricted to a discrete set of values [11]. KL divergence can capture this change and quantify the loss of information. In uniform quantization, where the quantization levels are uniformly distributed. Non-uniform quantization, which adapts the quantization levels to the actual distribution of weights, achieves lower KL divergence values for the same number of bits [10].

Interpreting KL divergence values in the context of quantization requires an understanding of the specifics of a particular architecture and task. Embedding layers in language models can tolerate surprisingly high KL divergence values without significant performance degradation, often due to redundancy in the representations [13].

From our perspective, the practical use of KL divergence in quantization tracking involves several strategies. Sequential quantization on a layer-by-layer basis allows us to identify critical layers with high KL divergence and apply finer quantization to them. Adaptive quantization could dynamically adjust the number of bits for each layer based on the target KL divergence. In mixed-precision quantization, KL divergence has the potential to be used as a criterion to decide the bitwidth for individual layers or individual channels. Monitoring KL divergence during quantization-aware training

(QAT) implies the possibility to optimize the learning process and prevent model degradation.

The advantages of KL divergence include its strong theoretical foundations in information theory [1], natural interpretation as a measure of information loss, in addition to the assumed ability to capture subtle differences in distributions.

From a different perspective, it also has significant limitations. The asymmetry of the KL divergence together with the requirement for accurate probability distributions can pose a challenge, but it also provides scope for more accurate and sensitive analyses even in the context of continuous weights [12] can be infinite if the distributions are not identical, and differences in the probabilities can lead to large changes in the divergence.

Also of interest is the possibility of using KL divergence in combination with other metrics mentioned in the above, i.e. cosine similarity or JS divergence for a more comprehensive view of changes in the distribution. This is especially the case when using the QAT mnethod [13]. It is important to properly treat numerical problems in the implementation, especially when dealing with logarithms of small numbers. For visualization, it appears useful to track the evolution of the KL divergence during training or sequential quantization. KL divergence represents a promising metric for comparing quantization methods. Due to its specific properties, it can efficiently capture changes in the distribution of neural network weights during the quantization process. In doing so, its ability to quantify information loss makes it a useful tool.

**Advantages:**

- KL divergence is very sensitive to small differences between the probability distributions being compared, allowing even subtle changes in model weights or in the behavior of data distributions to be accurately detected.
- The resulting value has a clear interpretation - it expresses the amount of information (in bits or nata) that would be "lost" if an approximating distribution Q were used instead of the true PP distribution. This property is valuable when evaluating models quantitatively or when optimizing model parameters.
- KL divergence has many practical uses - in our case, we expect it to be good at showing how one model mimics another, tracking changes in neural networks after they have been modified (e.g., after quantization), checking the progress of training models, and being able to detect unusual inputs that differ from the training data.
- It is the basis for deriving other additional metrics such as Jensen-Shannon divergence or various regularization techniques in variational autoencoders (VAEs).

**Limitations:**

- KL divergence is asymptotic. Therefore, it cannot serve as a full distance metric (it does not satisfy the symmetry metric), which may be limiting in some comparative analyses.

- The upper limit of KL divergence is not bounded (i.e., it can take arbitrarily high values), which makes direct interpretation of large values difficult and may cause numerical instability in extreme distributions.
- If there are values for which P(i)>0P(i)>0 and at the same time Q(i)=0Q(i)=0, the result is formally infinite (divergence is undefined), which is practically important when working with discrete distributions or in situations with limited data (e.g., in the case of rare events).
- In practice, the KL divergence can be sensitive to the choice of base distributions and to numerical errors due to rounding or missing probabilities. Therefore, it is often necessary to introduce additive smoothing or other techniques to stabilize the calculation.
- When comparing models with very different distributions, the result can be misleading or difficult to interpret, especially if the reference distribution Q does not capture all possible values observed in P.
- In the context of neural network quantization, this metric will be used to compare the distribution of weights before and after the quantization process. A high KL divergence value indicates that there has been a significant change in the distribution of weights, which can potentially lead to a degradation in model performance. This tool is therefore crucial in identifying problem areas in the quantization process. Regular monitoring of the KL divergence during the training process can provide information about the impact of quantization on each layer of the neural network. This allows us to identify potentially sensitive layers that may require strategy modification. This allows us to design methods for optimizing quantization processes, in order to maintain model performance even with reduced bit precision.

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Another metric that can be analyzed to assess the differences between the distributions of weights before and after quantization is Cosine Similarity.

**b) Cosine similarity**

It is a fundamental geometric metric that quantifies the similarity between two non-zero vectors in a multidimensional space by measuring the cosine of the angle between them [1]. This metric [2], has become an important tool in modern machine learning due to its ability to assess the directional similarity of vectors independent of their size. Its invariance and computational efficiency give it the capability for a wider range of applications from natural language processing (NLP) to neural network analysis. In the context of the project, it has the potential to be similar for comparing the weight vectors of individual layers or filters.

**Mathematical definition: for two vectors A and B, cosine similarity is defined as:**

$\cos(\theta) = (A \cdot B) / (||A||\ ||B||)$

where:

- $A \cdot B$ is the scalar product of the vectors
- $||A||$ a $||B||$ are Euclidean norms of vectors

**The cosine similarity results in:**

- The value is always in the interval [-1, 1]
- A value of 1 means identical direction of vectors (maximum similarity)
- A value of 0 means orthogonal (perpendicular) vectors
- Value -1 means opposite direction (maximum dissimilarity)

In the field of natural language processing, cosine similarity is the standard for measuring the semantic similarity of text documents [3]. When representing documents using vector models such as TF-IDF or modern word embeddings such as Word2Vec [4] or BERT [5], cosine similarity allows efficient comparisons between documents of different lengths without the distortion caused by word count. In computer vision, it is used to compare feature vectors extracted from convolutional neural networks [6], enabling tasks such as face recognition or similar image retrieval [7], [8].

Mathematically, the cosine similarity for two vectors A and B is defined as the normalized scalar product, which corresponds to the cosine of the angle θ between them. For vectors in n-dimensional space, it is computed as the ratio of the scalar product of the vectors and the product of their Euclidean norms [1]. The resulting value lies in the interval [-1, 1], where 1 indicates perfect direction congruence (zero angle), 0 indicates orthogonality (90° angle), and -1 expresses exactly opposite directions (180° angle). This normalization ensures that the metric is insensitive to the absolute size of the vectors.

$$\cos(\theta) = (A \cdot B) / (||A|| \cdot ||B||)$$

When applied to quantization analysis of neural networks, cosine similarity provides a different approach to comparing distributions than the other methods evaluated. While KL divergence measures changes in the probability distribution of the weights, cosine similarity captures the preservation of the relative relationships between the weights [9].

The advantages of cosine similarity include its computational efficiency with O(n) linear complexity, intuitive geometric interpretation, and broad support in all major machine learning frameworks [10]. The implementation is relatively simple and numerically stable for most practical applications. Moreover, cosine similarity is functional with sparse vectors, which is advantageous when dealing with high-dimensional representations.

Limitations of this metric :

Insensitivity to the size of vectors, although often advantageous, can mask important changes in the absolute values of the weights [11]. For vectors close to zero, numerical instability may occur, requiring special treatment in the implementation. Cosine similarity also fails to capture nonlinear relationships, which can be limiting when analyzing deep networks with complex interactions [12].

It is advisable to combine cosine similarity with complementary metrics for a holistic view of quantization effects. Combining with the L2 norm provides information on both magnitude and direction changes [13], while using it together with KL divergence allows capturing both distributional and directional changes. For visualization, it is efficient to generate heatmaps of cosine similarity between layers before and after quantization, allowing for quick identification of problematic areas of the model.

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### b) Jensen-Shannon divergence

It is a symmetric statistical metric that quantifies the similarity between two probability distributions [1]. Developed as a symmetric modification of the Kullback-Leibler divergence, the JS divergence addresses key limitations of its asymmetric predecessor and provides a robust, bounded measure of the dissimilarity of distributions [2]. This metric, has become an important tool in machine learning due to its mathematical contribution and practical properties, in particular its finiteness and symmetry.

Mathematically defined as:

$$JS(P||Q) = \tfrac{1}{2} \cdot D\_KL(P||M) + \tfrac{1}{2} \cdot D\_KL(Q||M)$$

In the context of machine learning, JS divergence finds wide application in critical domains. In training generative adversarial networks (GANs), it serves as a more stable alternative to standard loss functions, providing a gradient signal even when the distributions do not overlap [3]. In natural language processing, it is used to measure the similarity of word or topic distributions, which is crucial for tasks such as topic change detection or language model comparison [4]. In neural network compression, it can provide insight into changes in the distribution of weights, combining the benefits of KL divergence with additional symmetry and boundedness. In clustering analysis, it serves as a distance metric for hierarchical clustering of probability distributions [5], [6].

Mathematically, JS divergence is expressed as the weighted average of two KL divergences - the first between distribution P and the average distribution M, and the second between distribution Q and the same average distribution M. When applied to quantization of neural networks, JS divergence provides various advantages. Its symmetry means that it does not matter whether we consider the original weights as reference to the quantized ones or vice versa, which is conceptually simple and practically useful [7], [8].

Advantages of JS divergence include its mathematical robustness - it always provides finite values, which is critical when working with discretized weights [9]. Symmetry eliminates the need to decide on a reference distribution, simplifying interpretation. Boundedness allows direct comparison across different strata and models without normalization. Continuity and differentiability make JS divergence suitable for gradient optimization methods [10]. Moreover, JS divergence naturally interpolates between distributions, which is useful in stepwise quantization.

**Limitations of JS divergence:**

The computational complexity is higher than that of the simple KL divergence due to the need to compute the average distribution [11]. For high-dimensional distributions, it can be sensitive to the way the probability density is estimated. Also, JS divergence may not capture subtle local differences as well as some alternative metrics such as Wasserstein distance [12]. For extremely sparse distributions, it may provide overly optimistic values due to its tendency to average out differences.

Combining it with perceptual metrics provides a holistic view to preserve model functionality [13]. For visualization, it is effective to create distribution plots showing the original, quantized, and averaged distributions, allowing for an intuitive understanding of what JS divergence measures [14]. For implementation, it is critical to efficiently compute the average distribution, use numerically stable implementations of logarithms, and store intermediate results when recomputing. For large models, a sampling-based approximation of the JS divergence is recommended, which provides sufficient accuracy at a fraction of the computational cost [15].

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